MIL-S-19500/46B(EL)
25 APRIL 1966
SUPERSEDING
MIL-S-19500/46A(EL)
29 July 1963

MILITARY SPECIFICATION

TRANSISTOR, PNP, GERMANIUM, POWER
TYPES 2N574, 2N575, 2N575A, 2N1157A

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for germanium, PNP, transistors for relatively high-power circuit applications, and having the following particular characteristics at T_{MB} = + 25° ± 3°C. (See 3.2 herein.):

	h_{FE} (at: I_C = -10Adc, V_{CE} = -2Vdc)		h_{FE} (at: 1/ I_C = -25 Adc, V_{CE} = -2 Vdc)		h_{FE} (at: 1/ I_C = -40 Adc, V_{CE} = -2 Vdc)		$V_{CE(sat)}$ (at: I_C = -10 Adc, I_B = -2Adc)		$V_{CE(sat)}$ (at: 1/ I_C = -25Adc, I_B = -3.75Adc)		$V_{CE(sat)}$ (at: 1/ I_C = -40Adc, I_B = -6Adc)		θ_{J-MB}
	---		---		---		Vdc		Vdc		Vdc		°C/W
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
2N574	9	22	2/	2/	2/	2/	---	0.2	2/	2/	2/	2/	---	0.4
2N575	19	42	10	---	2/	2/	2/	2/	---	0.5	2/	2/	---	0.4
2N575A	19	42	10	---	2/	2/	2/	2/	---	0.5	2/	2/	---	0.4
2N1157A	38	84	2/	2/	10	---	2/	2/	2/	2/	---	0.8	---	0.4

1/ Pulsed: pulse width = 8.3 msec, duty cycle = 50%.

2/ Not applicable to this transistor type.

1.2 Maximum ratings.-

	P_C 1/	I_C	BV_{CEO}	BV_{CES}	V_{CBO}	V_{EBO}	T_J	T_{stg}	Altitude
	W	Adc	Vdc	Vdc	Vdc	Vdc	°C	°C	ft.
2N574	187	10	55	55	60	28	100	-65 to + 100	85,000
2N575	187	25	50	55	60	28	100	-65 to + 100	85,000
2N575A	187	25	55	65	80	28	100	-65 to + 100	85,000
2N1157A	187	40	50	60	80	28	100	-65 to + 100	85,000

1/ This collector power dissipation at T_{MB} = + 25°C. For collector power rating at T_{MB} = higher than + 25°C and up to + 100°C, derate at 2.5 W/°C.

FSC-5961

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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19500	Semiconductor Devices, General Specification For

STANDARDS

MILITARY

MIL-STD-750	Test Methods For Semiconductor Devices

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General.- Requirements for the transistors shall be in accordance with Specification MIL-S-19500, and as otherwise specified herein.

3.2 Abbreviations and Symbols.- The abbreviations and symbols used herein are defined in Specification MIL-S-19500 and as follows:

P_C - - - - - - - - - - - - collector power dissipation
T_{MB} - - - - - - - - - - - mounting-base temperature
θ_{J-MB} - - - - - - - - - - thermal resistance: junction to mounting base

3.3 Design and construction.- The transistors shall be of the design, construction, and physical dimensions specified in Figure 1 herein.

3.3.1 Terminal arrangement.- The terminal arrangement on the transistor shall be as indicated in Figure 1 herein.

3.3.2 Operating position.- The transistor shall be capable of proper operation in any position.

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3.4 Performance requirements.- The transistor performance characteristics shall be as specified in Tables I, II, and III herein. Except where specifically differentiated for respective transistor types (see 1.1, 1.2, and Tables I, II, and III herein), the performance requirements, including characteristics, ratings, and test conditions, apply equally to all transistor types covered herein.

3.5 Marking.- Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the 'classification' type designation only.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and Quality Conformance inspection.- Qualification and Quality Conformance inspection shall be in accordance with requirements in Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Groups A, B, and C inspection shall consist of the examinations and test specified in Tables I, II, and III, respectively, herein. Quality Conformance inspection shall include inspection of Preparation For Delivery (see 5.1).

4.2.1 Interval for End-Point Test measurements.- All applicable End-Point Test measurements shall be performed, after sample units have been subjected to required physical-mechanical or environmental test(s), in accordance with the following time-delay limitations:

(a) For Qualification inspection: within 16 hours.

(b) For Quality Conformance inspection: within 96 hours.

4.3 Group C testing.- Unless otherwise specified, Group C tests shall be performed on the initial lot and thereafter on a lot every 90 days or every fifth lot, whichever occurs first. (See Table III herein.) The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product(s) concerned. Upon determination by the Government inspector (in advance of Group C, 90-day or fifth-lot test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection (normally for a minimum of 3 consective lots) to be performed for required Group C tests.

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4.4 Specified LTPD for subgroups. - The LTPD specified for a subgroup in Tables I, II, and III herein shall apply for all of the tests, combined, in the subgroup.

4.5 Disposition of sample units. - Sample units that have been subjected to Group B, Subgroup 5 test shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroup 1, 2, 3, 4, 6, and 7 tests, and Group C Subgroup 1 test may be delivered on the contract or order provided that, after Groups B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.6 Particular examination and test procedures. -

4.6.1 Base-To-Emitter Voltage test. - The specified currents shall be applied to applicable terminals under the conditions specified (see 4.6.2 herein), and the base-to-emitter voltage shall then be measured. (The data in Standard MIL-STD-750, Method 3066, Test Cond. B, may be referred to as a guide for test procedure.)

4.6.2 Pulse conditions for particular tests. - To limit collector dissipation, the sample units subjected to the tests to which this paragraph is pertinently referenced shall be tested in the necessary circuit with a pulse: pulse width = 8.3 msec, duty cycle = 50%.

4.6.3 Mechanical damage resulting from tests. - Except for inherently deforming, multilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample as a result of any of the Group A or B or C tests.

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Table I. Group A inspection.

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		10				
2071	Visual and mechanical inspection	---		---	---	---	---
	Subgroup 2		10				
3036	Collector-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 60 Vdc I_E = 0		I_{CBO}	---	-7	mAdc
3036	Collector-to-base cutoff current:	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 80 Vdc I_E = 0					
	2N575A, 2N1157A			I_{CBO}	---	-20	mAdc
3061	Emitter-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{EB} = - 28 Vdc I_C = 0		I_{EBO}	---	-15	mAdc
3076	Static forward-current transfer ratio:	I_C = - 10 Adc V_{CE} = - 2 Vdc					
	2N574			h_{FE}	9	22	---
	2N575, 2N575A			h_{FE}	19	42	---
	2N1157A			h_{FE}	38	84	---
2/	Base-to-emitter voltage	I_C = 10 Adc V_{CE} = - 2 Vdc		V_{BE}	---	-1	Vdc
3020	Emitter floating potential:	R_{EB} = 10K ohms					
	2N574, 2N575,	V_{CB} = - 60 Vdc		V_{EBF}	---	-0.4	Vdc
	2N575A, 2N1157A	V_{CB} = -60 Vdc		V_{EBF}	---	-0.4	Vdc
	2N575A, 2N1157A	V_{CB} = - 80 Vdc		V_{EBF}	---	-0.5	Vdc

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Table I. Group A inspection-(Continued).

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3		10				
3011	Breakdown voltage, collector-to-emitter:	3/ Bias Cond. D I_C = 1 Adc I_B = 0					
	2N574, 2N575A			BV_{CEO}	55	---	Vdc
	2N575, 2N1157A			BV_{CEO}	50	---	Vdc
3011	Breakdown voltage, collector-to-emitter:	3/ Bias Cond. C I_C = 1 Adc I_B = 0					
	2N574, 2N575			BV_{CES}	55	---	Vdc
	2N575A			BV_{CES}	65	---	Vdc
	2N1157A			BV_{CES}	60	---	Vdc
3076	Static forward-current transfer ratio:	3/ I_C = - 25 Adc V_{CE} = - 2 Vdc					
	2N575, 2N575A			h_{FE}	10	---	---
3076	Static forward-current transfer ratio:	3/ I_C = - 40 Adc V_{CE} = - 2 Vdc					
	2N1157A			h_{FE}	10	---	---
2/	Base-to-emitter voltage:	3/ I_C = - 25 Adc V_{CE} = - 2 Vdc					
	2N575, 2N575A			V_{BE}	---	-2	Vdc
2/	Base-to-emitter voltage:	I_C = - 40 Adc V_{CE} = - 2 Vdc					
	2N1157A			V_{BE}	---	-2	Vdc

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Table I. Group A inspection-(Continued).

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3 -(Cont'd)						
3071	Saturation voltage, collector-to-emitter:	I_C = - 10 Adc I_B = - 2 Adc					
	2N574			$V_{CE(sat)}$	---	-0.2	Vdc
3071	Saturation voltage, collector-to-emitter:	3/ I_C = - 25 Adc I_B = - 3.75 Adc					
	2N575, 2N575A			$V_{CE(sat)}$	---	-0.5	Vdc
3071	Saturation voltage, collector-to-emitter:	3/ I_C = - 40 Adc I_B = - 6 Adc					
	2N1157A			$V_{CE(sat)}$	---	-0.8	Vdc
3061	Emitter-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{EB} = - 2 Vdc I_C = 0		I_{EBO}	---	-1	mAdc
3036	Collector-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 2 Vdc I_E = 0		I_{CBO}	---	-0.7	mAdc
	Subgroup 4		10				
3151	Thermal resistance, junction-to-mounting base	---		θ_{J-MB}	---	0.4	°C/W
3141	Thermal response time	T_{MB} = + 25° ± 3°C		τ_r	20	---	msec

1/ See 3.4 herein.

2/ See 4.6.1 herein.

3/ See 4.6.2 herein.

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Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		20				
2066	Physical dimensions	---		---	---	---	---
	Subgroup 2		15				
2031	Soldering heat	Immersion to within 1/8 ± 1/32 in. from case		---	---	---	---
1051	Temperature cycling	Test Cond. B except $T_{(high)} = +100° {}^{+3°}_{-0°}C$		---	---	---	---
1056	Thermal shock (glass strain)	Test Cond. A $T_{(high)} = +85° {}^{+5°}_{-0°}C$		---	---	---	---
1021	Moisture resistance	No initial conditioning		---	---	---	---
	End-point tests:						
3036	Collector-to-base cutoff current	Bias Cond. D $T_{MB} = +25° \pm 3°C$ $V_{CB} = -60$ Vdc $I_E = 0$		I_{CBO}	---	-11	mAdc
3036	Collector-to-base cutoff current:	Bias Cond. D $T_{MB} = +25° \pm 3°C$ $V_{CB} = -80$ Vdc $I_E = 0$					
	2N575A, 2N1157A			I_{CBO}	---	-30	mAdc
3061	Emitter-to-base cutoff current	Bias Cond. D $T_{MB} = +25° \pm 3°C$ $V_{EB} = -28$ Vdc $I_C = 0$		I_{EBO}	---	-23	mAdc
3076	Static forward-current transfer ratio:	$I_C = -10$ Adc $V_{CE} = -2$ Vdc					
	2N574			h_{FE}	6	24	---
	2N575, 2N575A			h_{FE}	12	48	---
	2N1157A			h_{FE}	25	100	---

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Table II. Group B inspection-(Continued).

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3		15				
2016	Shock	Non-operating G = 1500 5 blows of 0.5 msec ea. in orientations X1, Y1, Y2, Z2 (total = 20 blows)		---	---	---	---
2046	Vibration fatigue	Non-operating G = 20		---	---	---	---
2056	Vibration, variable frequency	---		---	---	---	---
2006	Constant acceleration (centrifuge)	G = 5,000 Orientations X1, Y1, Y2, Z2		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 2, above						
	Subgroup 4 2/		20				
3/	High-temperature operation: $T_A = +71° {}^{+3°}_{-0°}C$			---	---	---	---
3036	Collector-to-base cutoff current	Bias Cond. D $V_{CB} = -2$ Vdc $I_E = 0$		I_{CBO}	---	40	mAdc
3/	Low-temperature operation: $T_A = -55° {}^{+3°}_{-0°}C$			---	---	---	---
3076	Static forward-current transfer ratio:	$I_C = -10$ Adc $V_{CE} = -2$ Vdc					
	2N574			h_{FE}	9	---	---
	2N575, 2N575A			h_{FE}	19	---	---
	2N1157A			h_{FE}	38	---	---

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Table II. Group B inspection- (Continued).

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 4- 2/ (Cont'd)						
2036	Terminal torque	Test Cond. D Torque = 20 in. - oz. t = 15 sec.		---	---	---	---
2036	Tension	Test Cond. A Weight = 10 lb. ± 10 oz. t = 15 sec.		---	---	---	---
2036	Stud torque	Test Cond. D Torque = 20 lb. -in. t = 15 sec		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 2, above						
	Subgroup 5		20				
1046	Salt spray (corrosion)	Test Cond. B 4/		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 2, above						
	Subgroup 6		λ=10				
1031	High-temperature life (non-operating)	T_{stg} = + 100° $^{+3°C}_{-0°C}$		---	---	---	---
	End-point tests:						
3036	Collector-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 60 Vdc I_E = 0		I_{CBO}	---	-14	mAdc
3036	Collector-to-base cutoff current:	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 80 Vdc I_E = 0					
	2N575A, 2N1157A			I_{CBO}	---	-40	mAdc

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Table II. Group B inspection-(Continued).

Test Method per MIL-STD-750	Examination or test	Conditions 1/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 6-(Cont'd)						
3061	Emitter-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{EB} = - 28 Vdc I_C = 0		I_{EBO}	---	-30	mAdc
3076	Static forward-current transfer ratio:	I_C = - 10 Adc V_{CE} = - 2 Vdc					
	2N574			h_{FE}	6	24	---
	2N575, 2N575A			h_{FE}	12	48	---
	2N1157A			h_{FE}	25	100	--
	Subgroup 7		λ=20				
1026	Steady state operation life	T_{MB} = + 55° $^{+5°}_{-0°}$C I_C = 4 A V_{CE} = - 12 Vdc		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 6, above						

1/ See 3.4 herein.

2/ For this Subgroup, the sample units subjected to the High Temperature Operation test shall be permitted to return to and be stabilized at room temperature prior to their being subjected to the Low Temperature Operation test.

3/ Measurement shall be made after thermal equilibrium has been reached at the temperature specified.

4/ Marking shall have remained legible at conclusion of this test.

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Table III. Group C inspection. 1/

Test Method per MIL-STD-750	Examination or test	Conditions 2/	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		15				
1001	Barometric pressure, reduced (altitude operation):	Pressure = 15 ± 2 mmHg t = 1 min.					
3036	Collector-to-base cutoff current	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 60 Vdc I_E = 0		I_{CBO}	---	-7	mAdc
3036	Collector-to-base cutoff current:	Bias Cond. D T_{MB} = + 25° ± 3°C V_{CB} = - 80 Vdc I_E = 0					
	2N575A, 2N1157A			I_{CBO}	---	-20	mAdc

1/ Periodicity: See 4.3 herein.

2/ See 3.4 herein.

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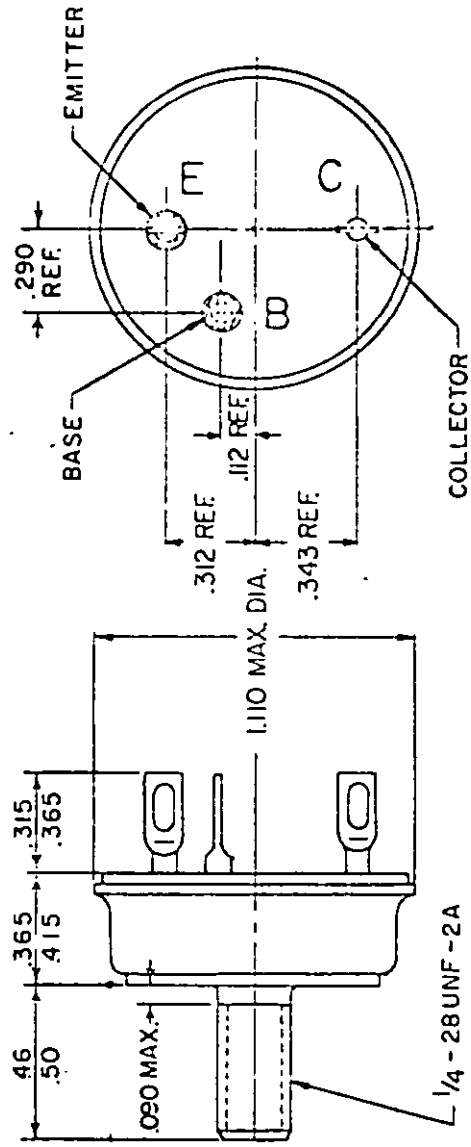


Figure 1. Outline and dimensions.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Unless otherwise specified in the bid request or contract, preparation for delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes includes in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this specification.

6.2 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, attention: AMSEL-PP-EM-2.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify):*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Includ Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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